C9.6 Transaction requirements

This section summarizes the requirements of transactions associated with Exclusive accesses.

The existing AXI rules apply for all transactions with **AxDOMAIN** set to System Shareable.

For transactions with **ARDOMAIN** set to Inner Shareable or Outer Shareable:

- an Exclusive Load transaction must assert **ARLOCK**
- an Exclusive Load transaction must use either ReadClean or ReadShared transaction type

  Note

  It is not required that a transaction is issued for the execution of a `LDREX` instruction.

- any slave that is capable of supporting Exclusive transactions must give an EXOKAY response to an Exclusive Load transaction

  Note

  An OKAY response to an Exclusive Load transaction indicates that Exclusive transactions are not supported to that address location and all Exclusive Store transactions to that location will also return an OKAY response.

- an Exclusive Store transaction must assert **ARLOCK**
- an Exclusive Store transaction must be a CleanUnique transaction
- an Exclusive Store transaction response can be either EXOKAY or OKAY
- matching Exclusive Load transactions and Exclusive Store transactions are not required.

  Note

  An Exclusive Load transaction can occur with no matching Exclusive Store transaction. An Exclusive Store transaction can occur with no matching Exclusive Load transaction.

When multiple Exclusive-capable threads use a single interface:

- transactions must use an AXI ID value that permits the Exclusive-capable thread to be identified
- the bits of the AXI ID signal that are used to identify the Exclusive-capable thread must be the same for all Exclusive transactions from the same Exclusive-capable thread.

A single Exclusive-capable thread must not have an Exclusive Store transaction in progress at the same time as any transaction that registers that a master is performing an Exclusive sequence.

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Chapter C10
Optional External Snoop Filtering

This chapter describes using an external snoop filter with an existing master component. It contains the following sections:

- *About external snoop filtering* on page C10-270
- *Master requirements to support snoop filters* on page C10-272
- *External snoop filter requirements* on page C10-273.

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